

Ibrahim Yalcin

Senior Software Engineer · AI-Native Platform Engineering

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Senior full-stack software engineer with **7+ years** building mission-critical payroll, HR and pensions systems for UK schools and Multi-Academy Trusts. One of two senior engineers with full technical ownership of Juniper Education's Catalyst payroll platform through a major organisational restructure and a live third-party (PeopleFirst) migration; following the decision to move off that dependency, now co-building Juniper's own in-house payroll & HR platform and contributing to an AI-powered school-insights product. Independently designed, built and shipped a CFO-commissioned production integration end-to-end using agentic AI tooling (Claude Code, MCP servers). Works day-to-day in Clean/Onion Architecture, DDD, CQRS, multi-tenant database isolation, and cloud deployment across Azure (AKS) and AWS. MSc Financial Engineering (93% GPA) adds a quantitative edge that suits regulated, numbers-heavy domains like payroll.

EXPERIENCE

Juniper Education

Senior Software Engineer

Apr 2024 – Present

One of two senior engineers retained during a major organisational restructure, sharing full ownership of the technical roadmap, third-party migrations, and the production platform.

- Designed and co-led the live integration of PeopleFirst as the new payroll calculation engine, maintaining backward compatibility across three simultaneous migration states (legacy-only, hybrid, fully migrated) in a single production codebase.
- Helped identify and close a critical reporting gap: PeopleFirst provided no LGPS pension returns, no BACS file generation, no payroll reporting. Contributed to a dedicated reporting layer consuming PeopleFirst data, protecting all existing client contracts post-migration.
- Contributed to a parallel-run quality gate that compared old and new system outputs byte-for-byte before each school cutover, catching zero financial discrepancies across all migrated schools.
- Helped maintain a three-tier multi-tenant database architecture (central registry, per-school SQL Server databases, and MAT shared databases) serving hundreds of isolated client datasets under GDPR.
- Part of the team responsible for 50+ bespoke LGPS pension fund report templates for individual UK council pension schemes (Bexley, Croydon, Dorset, Surrey, Gloucestershire, Kent, Wiltshire, Suffolk).
- Co-building Juniper's own in-house payroll & HR platform with an HMRC-spec-compliant tax/NI engine, using an AI-native workflow (Claude Code, MCP servers) for implementation, migration tooling and tests.
- Contributing engineer on Juniper Intelligence, an Anthropic Claude-powered product giving school staff conversational access to attendance and assessment insights.

INDEPENDENT PROJECT · CFO-COMMISSIONED

SugarCRM / Zendesk integration, built solo end-to-end using Claude Code: a .NET 8 worker with watermark-based sync and resilient API handling, deployed on AWS (ECS/EventBridge) with 298 automated tests.

Software Engineer — .NET Payroll

Dec 2021 – Apr 2024

- Part of the team that rebuilt the core payroll platform from a legacy codebase using Onion Architecture alongside the legacy layer, applying a strangler fig pattern to migrate features incrementally without disrupting live payroll for hundreds of schools.
- Contributed to the per-tenant database resolution layer: a runtime resolver reads the central school registry, fetches server credentials from Azure Key Vault via Managed Identity, and assembles a per-school connection string at runtime, so no secrets ever touch disk.
- Helped architect the Application layer as vertical feature slices (Bank Accounts, Contracts, Employees, Back Pay, Pension Schemes) using typed request/response models, CQRS application services, and functional result types in place of exceptions.
- Contributed to the background worker scheduling engine: a .NET BackgroundService with a 60-second database-polling loop, a priority queue with jump-queue override, a master kill switch for zero-redeployment shutdowns, and 10+ typed processors covering LGPS, BACS, RTI, payslip email, approval, and queue processing.
- Helped establish Gitflow branching and Azure DevOps PR-driven state transitions for the team.

Paragon Education Cambodia

Data Scientist

2019 – Aug 2021

- Applied Python (Pandas, NumPy) and statistical analysis to build financial forecasting, reporting, and data-analysis tooling for school group leadership, alongside graduate study in Financial Engineering.
- Analysed financial and operational datasets to support data-driven decision-making across a Kindergarten-to-university international school group.

Mathematics Teacher

2014 – 2019

- Taught A-Level Maths and Statistics in the high school section of a Kindergarten-to-university international school group before moving into data science.

PERSONAL PROJECTS

TakeawaySaaS — Multi-Tenant Takeaway Ordering Platform

2026 – Present

- Designed and built a multi-tenant SaaS platform for takeaway ordering using an AI-native workflow (Claude Code): .NET 10, Blazor, Next.js 15, PostgreSQL and Stripe Connect. Real payments proven end-to-end; 950+ commits over 5 months.

EDUCATION

Graduate Certificate in Applied AI Fundamentals

In Progress

WorldQuant University

MSc Financial Engineering 93% GPA

WorldQuant University · 2019–2021 · USA

Thesis: PCA of the US Treasury Yield Curve

BSc Mathematics

Ataturk University · 2007–2011 · Turkey

KEY SKILLS

LANGUAGES

C# SQL/T-SQL Python TypeScript JavaScript R

FRAMEWORKS

ASP.NET Core React Angular Vue.js

Entity Framework Core EPPlus

DATA

SQL Server PostgreSQL SQLite Redis

CLOUD & DEVOPS

Azure (AKS, Key Vault, Service Bus) AWS (ECS, EventBridge)

Docker Azure DevOps GitHub Actions CI/CD

ARCHITECTURE

Clean/Onion Architecture DDD CQRS Multi-Tenant

Microservices RESTful APIs

AI-NATIVE ENGINEERING

Claude Code MCP servers Antigravity Codex

JetBrains AIR JetBrains Junie

DATA & ML

NumPy Pandas scikit-learn TensorFlow Keras

TESTING & DELIVERY

xUnit NUnit TDD Playwright Kanban/Scrum

Jira/Confluence

CERTIFICATIONS

TensorFlow: Deployment

TensorFlow Developer

IBM Applied AI

WQU Applied Data Science

IBM Data Science Professional

Google IT Support

IBM Cybersecurity Analyst